

Paul Pham

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EDUCATION

Washington University in St. Louis – St. Louis, MO

Expected Graduation: May 2029

Bachelor of Science in Electrical Engineering

GPA: 3.84

Minors: Computer Science, Philosophy | Dean's List: Fall 2025, Spring 2026

Relevant Coursework: Digital Logic and Computer Design, Differential Equations and Dynamical Systems Modeling, Linear Algebra and Component Analysis, Data Structures and Algorithms

TECHNICAL SKILLS

- **Hardware & Embedded:** KiCad (Schematic Capture & Custom PCB Design), CircuitPython, SystemVerilog, FPGA, Raspberry Pi, ESP32, Soldering & Electrical Instrumentation
- **Robotics & Control:** Inverse Kinematics, PID Control Loops, Computer Vision (OpenCV), ArUco Marker-Based Localization
- **Software & Tools:** MATLAB, Python, Java, Git & GitHub, CAD (Fusion 360), VS Code

TEAM EXPERIENCE

WashU Rocketry Team | Airbrakes

Fall 2026 – Present

- Designed the Airbrakes Compute Element (ACE), a custom PCB integrating an RP2040 microcontroller with a 9-DOF IMU and barometric altimeter for real-time rocket-state sensing during flight.
- Implemented a 6-DOF apogee prediction algorithm using 4th-order Runge-Kutta integration at 1Hz, feeding a PID controller that actuates a 4-paddle airbrake mechanism to hold apogee within a 4,000-6,000 ft target range.

WashU Robotics | Delivery

Fall 2026 – Present

- Designed a vibration-proof 22.2V LiPo power distribution system for a 40-lb delivery robot, isolating 24V drive, 7.4V servo, and 5V/10A compute rails behind a 40A fuse, XT90-S anti-spark circuit, and marine-grade bus bars.
- Engineered a serial protocol between a Raspberry Pi and Arduino motor controller, translating quadrature-encoder pulses into real-time RPM telemetry and velocity/position data for robotic odometry.
- Integrated a RealSense 435i depth camera, RPLidar, and ZED-F9P RTK GPS module onto a Raspberry Pi 5 compute stack to support autonomous navigation.

PROJECT EXPERIENCE

Autonomous Arm w/ Computer Vision (AACV) | Co-Developer

June 2026 – August 2026

- Co-engineered a 6-DOF robotic arm requiring autonomous object detection with no manual control; built an OpenCV/ArUco-based vision pipeline in Python that applies per-color HSV masking to localize 6 colored cubes, converting camera input into real-world coordinates calibrated to sub-centimeter accuracy off a single 24.75mm marker.
- Derived and coded an inverse-kinematics solver to convert target coordinates into 6 servo-joint commands in real time, enabling 2 fully autonomous modes: rainbow-order stacking and color-to-marker matching; built a two-layer calibration system (per-servo angle mapping plus a 3x3 interpolated correction grid) to reduce placement error.

Pomodoro Cube | Developer

June 2026 – August 2026

- Designed and hand-soldered a custom KiCad PCB for an ESP32-based device, CAD-modeling an enclosure and magnetic charging dock to fit every component within the 2.2-inch dimensions of a Rubik's Cube.
- Solved a 4-pin hardware I/O constraint with a resistor-ladder circuit multiplexing 4 buttons onto a single analog input; programmed CircuitPython firmware as a finite-state machine to run 3 interactive modes without memory fragmentation.

ACTIVITIES & LEADERSHIP

Vietnamese Student Association (VSA) | Outreach Chair

- Serve as Outreach Chair, personally contacting and negotiating with local and global businesses to secure \$1,000+ in sponsorships for VSA's annual Formal/cultural event, attended by 400+ students.

Lunar New Year Festival Exec Board | Multimedia Chair

- Edited 11 act intros and performances for WashU's annual Lunar New Year Festival show (650+ attendees), coordinating directly with each act's coordinators to translate their creative vision into final videos.